

Ahlam Abuawad
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Date of Preparation

July 26, 2023

ACADEMIC APPOINTMENTS / WORK EXPERIENCE

- 2022 - Present **Geisel School of Medicine at Dartmouth, Hanover, NH**
Postdoctoral Fellow
- 2017 - 2022 **Columbia University Mailman School of Public Health, New York, NY**
Graduate Researcher/PhD Candidate
Developed expertise in arsenic exposure, toxicity, adverse health-related outcomes and methods of intervention and prevention primarily focused on the role of nutrition on arsenic-related toxicity and, in turn, metabolic outcomes such as BMI and diabetes in Bangladeshi populations.
- 2017 **Collegiate Science & Technology Entry Program, Albany, NY**
University at Albany, State University of New York
Academic Adviser
Provided secondary academic counseling to underrepresented and minority students in the Science, Technology, Engineering, and Math (S.T.E.M.) and licensed professional fields, as well as supervised tutors, mentors, and support staff for efficiency in daily tasks.
- 2015 - 2017 **Collegiate Science & Technology Entry Program, Albany, NY**
University at Albany, State University of New York
Graduate Assistant
Developed and executed programs to promote student communication and excellence in their personal and academic lives as well as assisted students with cover letters, resumes, personal statements, and time management skills.
- 2015 - 2016 **University at Albany, School of Public Health, Rensselaer, NY**
Senior Research Assistant
Analyzed data for trends regarding the contaminant and nutrient concentrations of farmed and wild salmon.
- 2014 - 2015 **College of Nanoscale Science and Engineering, Albany, NY**
Research Assistant
Performed enhanced dark field and light field microscopy to identify the biodistribution of nanoparticles in rat models.
- 2013 - 2014 **Pharmaceutical Research Institute, Rensselaer, NY**
Research Assistant
Monitored the influence of anticoagulants on the clotting time and strength of plasma and blood using aPTT and TEG assays, respectively.
- 2012 - 2015 **Collegiate Science & Technology Entry Program, Albany, NY**
University at Albany, State University of New York
Tutor/Mentor
Assisted students with introductory biology, introductory chemistry, organic chemistry, and genetics coursework, and additionally guided students with academic issues as they attempted to acclimate to the collegiate environment

EDUCATION

- 2017 - 2022 **Columbia University Mailman School of Public Health**, New York, NY
Ph.D. in Environmental Health Sciences, Oct 2022
Advisor: Dr. Ana Navas-Acien
- 2017 - 2020 **Columbia University Mailman School of Public Health**, New York, NY
MPhil in Environmental Health Sciences, Oct 2020
- 2015 - 2017 **University at Albany, School of Public Health**, Albany, NY
MPH - Concentration in Environmental Health Sciences, May 2017
- 2011 - 2015 **University at Albany, State University of New York**, Albany, NY
B.S. in Biology, Summa Cum Laude, May 2015
B.A. in Chemistry, May 2015
Minor: Mathematics
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HONORS AND AWARDS

- 2023 International Society for Environmental Epidemiologists (ISEE) North American Chapter (NAC) Travel Awardee
- 2023 Geisel School of Medicine at Dartmouth Department of Epidemiology Travel Awardee
- 2021 ISEE 2021 Registration Awardee
- 2020 ISEE 2020 Virtual Awardee
- 2019 ISEE 2019 Travel Awardee
- 2017 - 2020 Allan Rosenfield Scholar
- 2017 William Randolph Hearst Fellowship Recipient
- 2017 President's Award for Leadership: Academic Leadership Award
- 2015 23rd Annual CSTEP Conference First Place Oral Presentation in Technology
- 2013 - 2015 Dr. Seth W. Spellman Jr. Academic Achievement Award
- 2012 Be Semi: Afro-Latino Scholarship Recipient (MALIK Fraternity, Inc.)
- 2012 President's Award for Leadership: Great Dane Award
- 2011 - 2015 Dean's List Recipient
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PROFESSIONAL ORGANIZATIONS, SOCIETIES AND SERVICE

Society Membership:

- 2023 - Present Society for Epidemiologic Research (SER) Member
- 2018 - Present ISEE Member
- 2017 - 2018 American Public Health Association (APHA) Member
- 2017 Delta Omega Public Health Honor Society, Alpha Gamma Chapter Inductee
- 2014 - 2015 Corps Historian, Five Quad VAS
- 2011 - 2015 Collegiate Science & Technology Entry Program (CSTEP) Member
- 2011 - 2015 Five Quad Volunteer Ambulance Service (VAS) Member

2011 - 2015 Presidential Honor Society Member

Peer Review Activities:

Environ Intern (Oct 18); *Environ Health Perspect* (Jan 20, Feb 20); *Environ Res* (Sept 20); *Environ Health* (May 21); *Environ Health* (Sept 21); *Environ Res* (Apr 23); *Environ Health* (Apr 23), *Environ Res* (Jul 23).

FELLOWSHIP AND GRANT SUPPORT

Active grant

1. Title of grant: "Training Program for Quantitative Biomedical Sciences in Cancer"
Dates: 11/01/2022-10/31/2023
Sponsoring agency: NCI 5T32CA134286-13
Principal investigator: Karagas/Tosteson/Whitfield
Funding level: US \$385,672
The Training Program for Quantitative Biomedical Sciences in Cancer at provides fellows the opportunity to participate in a combination of structured, group learning activities and individually designed mentored research opportunities.
Role: Postdoctoral fellow

Past grants

1. Title of grant: "Arsenic Methylation, One-Carbon Metabolism, and Diabetes Incidence"
Dates: 09/01/2020-08/31/2022
Sponsoring agency: NIEHS 1 F31 ES032321-01
Principal investigator: Ahlam Abuawad
Funding level: US \$67,136
Study addresses the interplay between modifiable, e.g. nutritional status, and environmental exposures such as arsenic, with diabetes and diabetes-related outcomes by investigating the influence of nutritional status on the association between arsenic methylation and diabetes.
Role: Principal investigator
 2. Title of grant: "KC Donnelly Externship - Promotion of Translational/Transdisciplinary Efforts in Graduate and Post-Doctoral Research"
Dates: 10/01/2021-12/31/2021
Sponsoring agency: NIEHS P42 ES010349
Principal investigator: Ahlam Abuawad
Funding level: US \$10,000
Study will provide valuable training in metabolomics, prenatal environmental exposures, disease epidemiology, and allow for collaboration with an interdisciplinary team of researchers that are experts in early-life toxic metal and metalloid exposures.
Role: Principal investigator
 3. Title of grant: "IMSD at Columbia's Mailman School of Public Health"
Dates: 09/01/2017-8/31/2020
Sponsoring agency: NIGMS R25GM062454
Principal investigator: Goldsmith/Martins/Navas-Acien/Wingood
Funding level: US \$504,694
The Initiative for Maximizing Student Development (IMSD) at Columbia's Mailman School of Public Health provides an education project for under-represented students that includes various strategies to address barriers in pursuing and successfully completing graduate-level education.
Role: Fellow
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TEACHING EXPERIENCE AND RESPONSIBILITIES

Specific Courses:

Spring 2020 Analysis of Environmental Health Data (33 enrolled students)
Spring 2020 Analysis of Environmental Health Data (35 enrolled students)
Fall 2019 Frameworks of Environmental Health Policy (10 enrolled students)
Fall 2018 Frameworks of Environmental Health Policy (18 enrolled students)

General Teaching Activities:

Summer 2021 Workshop Guide, Columbia Summer Research Institute (“Environmental Mixtures”)
Summer 2020 Workshop Guide, Columbia Summer Research Institute (“Environmental Mixtures”)
Summer 2019 Workshop Guide, Columbia Summer Research Institute (“Environmental Mixtures”)
2018 - Present Student Mentor/Editor, Brotherhood/Sister Sol
Summer 2018 Workshop Guide, Columbia Summer Research Institute (“Environmental Mixtures”)

PUBLICATIONS

Abuawad A, Zhao J, Domingo-Relloso A, Goldsmith J, Albrecht S, Cole S, ... Sanchez T. (2023). Arsenic methylation, one-carbon metabolism, and diabetes incidence in the Strong Heart Family Study: An untargeted metabolomics approach. (in preparation).

Abuawad A, Hoen A, Navas-Acien A, Gamble MV, and Karagas M. (2023). Early Life Arsenic Exposure, Metabolomics, and Gestational Diabetes Mellitus. (in preparation).

Jiang E, Domingo-Relloso A, **Abuawad A**, Haack K, Tellez-Plaza M, Fallin MD, ... Navas-Acien A. (2023). Arsenic exposure and accelerated aging: The association with fatal and nonfatal cardiovascular disease and all-cause mortality in the Strong Heart Study. (in review).

Abuawad A*, Bozack A*, Navas-Acien A, Goldsmith J, Liu X, Hall MN, ... Gamble MV. (2023). The Folic Acid and Creatine Trial: Treatment Effects of Supplementation on Arsenic Methylation Indices and Metabolite Concentrations in Blood in a Bangladeshi Population. *Environ. Health Perspect.* 131(3):37015. doi: 10.1289/EHP11270. Epub 2023 Mar 28. PMID: 36976258; PMCID: PMC10045040. (*co-first authors).

Abuawad A, Goldsmith J, Herbstman J, Parvez F, Islam T, Lolacono N, ... Gamble MV. (2022). Urine dilution correction methods utilizing urine creatinine or specific gravity in arsenic analyses: Comparisons to blood and water arsenic in the FACT and FOX studies in Bangladesh. *Water.* 14(9):1477. <https://doi.org/10.3390/w14091477>

Abuawad A, Bozack AK, Saxena R, Gamble MV. (2021). Nutrition, one-carbon metabolism and arsenic methylation. *Toxicology.* 15(457):152803. doi: 10.1016/j.tox.2021.152803. Epub 2021 Jun 15. PMID: 33905762; PMCID: PMC8349595.

Abuawad A, Spratlen MJ, Parvez F, Slavkovich V, Ilievski V, Lomax-Luu A, ... Gamble MV. (2021). Association between body mass index and arsenic methylation in three studies of Bangladeshi adults and adolescents. *Environ. Int.* 149:106401. doi: 10.1016/j.envint.2021.106401. Epub 2021 Apr. PMID: 33549917; PMCID: PMC7976732.

Navas-Acien A, Spratlen MJ, **Abuawad A**, Lolacono NJ, Bozack AK, Gamble MV. (2019). Early-Life Arsenic Exposure, Nutritional Status, and Adult Diabetes Risk. *Curr. Diab. Rep.* 19(12): 147. doi: 10.1007/s11892-019-1272-9. PMID: 31758285; PMCID: PMC7004311.

Gibson EA, Nunez Y, **Abuawad A**, Zota AR, Renzetti S., Devick KL, ... Kioumourtzoglou MA. (2019). An overview of methods to address distinct research questions on environmental

mixtures: an application to persistent organic pollutants and leukocyte telomere length. Environmental health: a global access science source, 18(1): 76. doi: 10.1186/s12940-019-0515-1. PMID: 31462251; PMCID: PMC6714427.

Abalu O, **Abuawad A**, González-Dávila M, Kingsbury D, Puac-Polanco V, Santaella J, Segura L. AJPH Global News. Am J Public Health. <https://ajph.aphapublications.org/>. December 2018 – Present.

Muralidharan-Chari, V., Kim, J., **Abuawad, A.**, Naeem, M., Cui, H., & Mousa, S. A. (2016). "Thymoquinone modulates blood coagulation in vitro via its effects on inflammatory and coagulation pathways." Int. J. Mol. Sci. 17(4): 474. Muralidharan-Chari V, Kim J, Abuawad A, Naeem M, Cui H, Mousa SA. Thymoquinone Modulates Blood Coagulation in Vitro via Its Effects on Inflammatory and Coagulation Pathways. Int J Mol Sci. 2016 Mar 30;17(4):474. doi: 10.3390/ijms17040474. PMID: 27043539; PMCID: PMC4848930.

PRESENTATIONS

- "Early life arsenic exposure, metabolomics, and gestational diabetes mellitus." International Society of Environmental Epidemiologists - North American Chapter Annual Meeting, Oregon, United States. (06/2023).
- "The folic acid and creatine trial: Treatment effects on arsenic metabolite concentrations in blood." International Society of Environmental Epidemiologists Annual Meeting, Athens, Greece. (09/2022).
- "Lipid metabolites and arsenic methylation in the Strong Heart Family Study: An untargeted metabolomics approach." International Society of Environmental Epidemiologists Annual Meeting, Athens, Greece. (09/2022).
- "Arsenic, nutrition, and metabolic outcomes." Columbia University Mailman School of Public Health Department of Environmental Health Sciences, New York, United States. (07/2022).
- "The role of nutrition and adiposity in arsenic methylation." Columbia University Mailman School of Public Health Department of Environmental Health Sciences, New York, United States. (04/2022).
- "Early life arsenic exposure, metabolomics, and gestational diabetes mellitus." Superfund Research Program Annual Meeting: KC Donnelly Externship, Virtual. (12/2021).
- "The role of nutrition and adiposity in arsenic methylation: Evidence from Bangladesh." Columbia University Mailman School of Public Health Department of Environmental Health Sciences, New York, United States. (04/2021).
- "Methods for accounting for urine dilution in arsenic analyses: comparisons to blood arsenic." Superfund Research Program Annual Meeting, Virtual. (12/2020).
- "Association between body mass index and arsenic methylation in three studies of Bangladeshi adults and adolescents." Superfund Research Program Annual Meeting, Virtual. (12/2020).
- "Methods for accounting for urine dilution in arsenic analyses: comparisons to blood arsenic." International Society of Environmental Epidemiologists Annual Meeting, Virtual. (08/2020).
- "The folic acid and creatine trial: Treatment effects on arsenic metabolite concentrations measured in blood." The Folic Acid, Vitamin B12, and One-Carbon Metabolism Conference, Virtual. (08/2020).
- "Arsenic assessment and epigenetics in diverse populations." Columbia University Mailman School of Public Health Department of Environmental Health Sciences, New York, United States. (04/2020).
- "Body mass index is associated with arsenic metabolism in three studies in Bangladesh." International Society of Environmental Epidemiologists Annual Meeting, Utrecht, Netherlands. (08/2019).

- “Body mass index is associated with arsenic metabolism in three studies in Bangladesh.” Columbia University Mailman School of Public Health Department of Environmental Health Sciences, New York, United States. (04/2019).
- “Body mass index is associated with arsenic metabolites in three studies in Bangladesh.” Northeast Regional Superfund Research Program Meeting, Massachusetts, United States. (4/2019).
- “Body mass index is associated with arsenic metabolites (in urine and blood) in the Bangladesh FACT study.” Superfund Research Program Annual Meeting, California, United States. (11/2018).
- “The association of urine and blood arsenic metabolites with body mass index in the Bangladesh FACT study.” International Society of Environmental Epidemiologists Annual Meeting, Ottawa, Canada. (08/2018).
- “The association of body mass index with arsenic metabolites (in urine and blood) in the Bangladesh FACT study.” Student Research Diversity Day, New York, United States. (04/2018).
- “The association of body mass index with arsenic metabolites (in urine and blood) in the Bangladesh FACT study.” Columbia University Mailman School of Public Health Department of Environmental Health Sciences, New York, United States. (04/2018).
- “Do farmed salmon have a more beneficial balance of nutrients and contaminants than wild salmon?” Annual University at Albany School of Public Health Student Poster Day, New York, United States. (05/2017).
- “NanoHealth & Safety Center: Investigation of nanoparticle biodistribution following inhalation exposure.” Annual Collegiate Science & Technology Entry Program Statewide Student Conference, New York, United States. (04/2015).
- “NanoHealth & Safety Center: Investigation of nanoparticle biodistribution following inhalation exposure.” Annual University at Buffalo Undergraduate Research Conference, New York, United States. (07/2014).
- “Anticoagulant properties of Thymoquinone extracted from black cumin seeds.” Annual Biology Undergraduate Research Symposium, New York, United States. (04/2014).
- “Anticoagulant properties of Thymoquinone extracted from black seed.” Annual Collegiate Science & Technology Entry Program Statewide Student Conference, New York, United States. (04/2014).
- “Anticoagulant properties of Thymoquinone extracted from black seed.” Annual Albany College of Pharmacy and Health Sciences Research Forum, New York, United States. (01/2014).
- “Comparative anticoagulant efficacy and safety among different approved and emerging anticoagulant therapies.” Annual University at Buffalo Undergraduate Research Conference, New York, United States. (07/2013).

VOLUNTEER EXPERIENCE

2019 - 2022	Riverside Park Conservancy , New York, NY Volunteer
2018 - 2022	American Journal of Public Health Global News , New York, NY Author
2018 - 2022	Brotherhood/Sister Sol , New York, NY Mentor
2011 - 2015	Five Quad Volunteer Ambulance Service , Albany, NY Driver/Emergency Responder/CPR Instructor/Corps Historian <i>Performed first aid procedures such as splinting, recording vital signs, back boarding, stair chair, Reeves, etc.</i>

SKILLS

Computer: Proficient in Microsoft Office Suite, SAS, STATA, and R

Language: Knowledge of basic Arabic and Spanish

Medical: CPR and AED instructor; First Aid, and EMT certifications (expired)
